Can AI Foundation Models Beat Traditional Statistical Models?

Forecasting daily demand for Walmart food items

GROUP 3

Charlotte Zhu · Shihan Wang · Jing Chang · Rachel Liu

Time Series Analysis and Forecasting · Dr. William Yu · October 22, 2026

Retail forecasting has to balance accuracy, efficiency and business value

TODAY'S DEFAULT

LightGBM

Machine learning; dominates this kind of retail data and has become the industry default

THE CHALLENGER

AI foundation models

Pre-trained on huge datasets; forecast a product they have never seen, with no training at all

So we ask:

1 Can pre-trained AI forecast as well as traditional methods, and faster?

2 Does a better forecast mean a better replenishment decision?

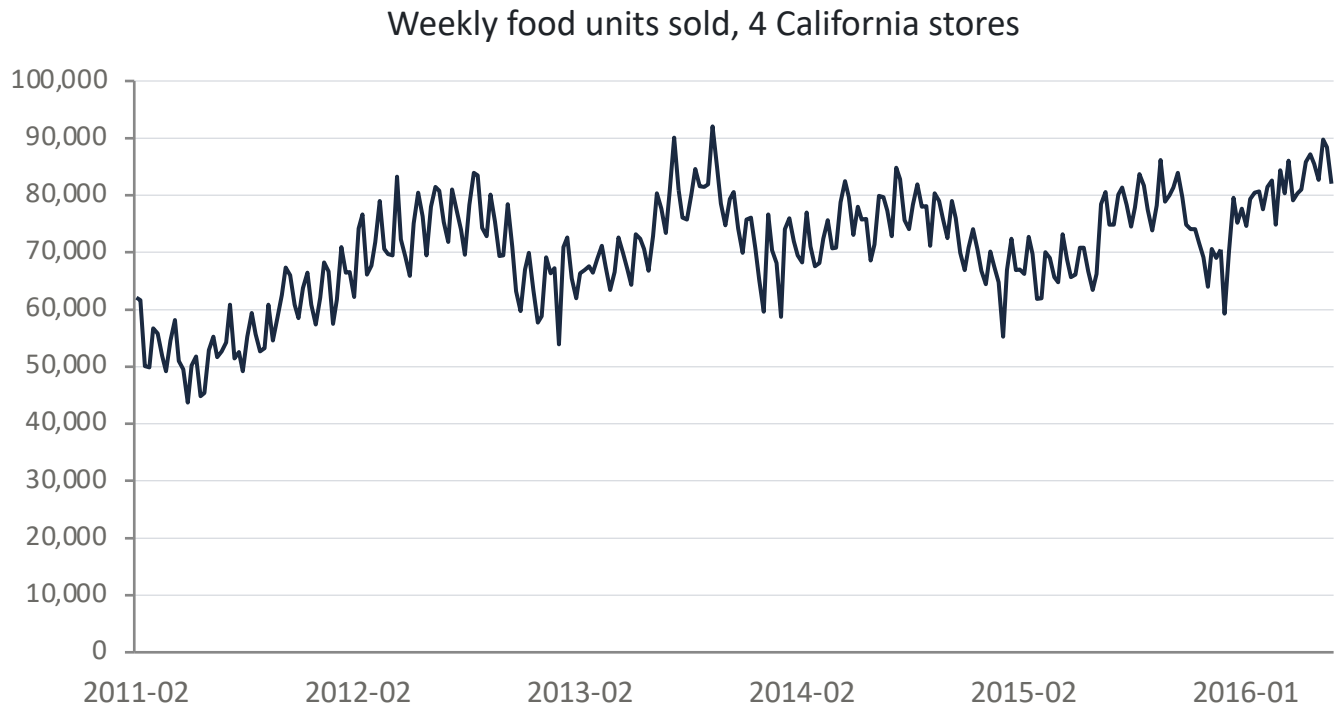


5,748 daily series, one 28-day test window

Training: Jan 29, 2011 – May 22, 2016 (1,941 days) · models fitted and tuned here

Test: 28 days

May 23 – Jun 19, 2016, never seen by any model. LightGBM tuning used only the last 28 training days.



4 stores × 1,437 items

California, food category = 5,748 item × store series

WRMSSE

The official M5 score: scaled, sales-weighted error; below 1 beats naive

One test window

All results come from a single 28 days; rankings could shift

Five models plus a simple average

Model	Family	How it is fitted	What it sees
AutoETS	Statistics	One model per series (exponential smoothing)	Own history
AutoARIMA	Statistics	One model per series (seasonal ARIMA)	Own history
LightGBM	Machine learning	One model trained across all items, predicts each item	History + price, calendar, SNAP
Chronos	AI foundation model	Pre-trained by Amazon, zero-shot: no training on our data	Own history
Ensemble	Combination	Unweighted average of AutoETS and LightGBM	Their two forecasts
Seasonal naive	Benchmark	Repeats the same weekday last week	Last week
Claude (experiment)	General-purpose AI	Reads sales as text and reasons; 112 series only	History + price, calendar

Inputs differ: AutoETS, AutoARIMA and Chronos see only each series' own history; LightGBM and Claude also see prices and the calendar.

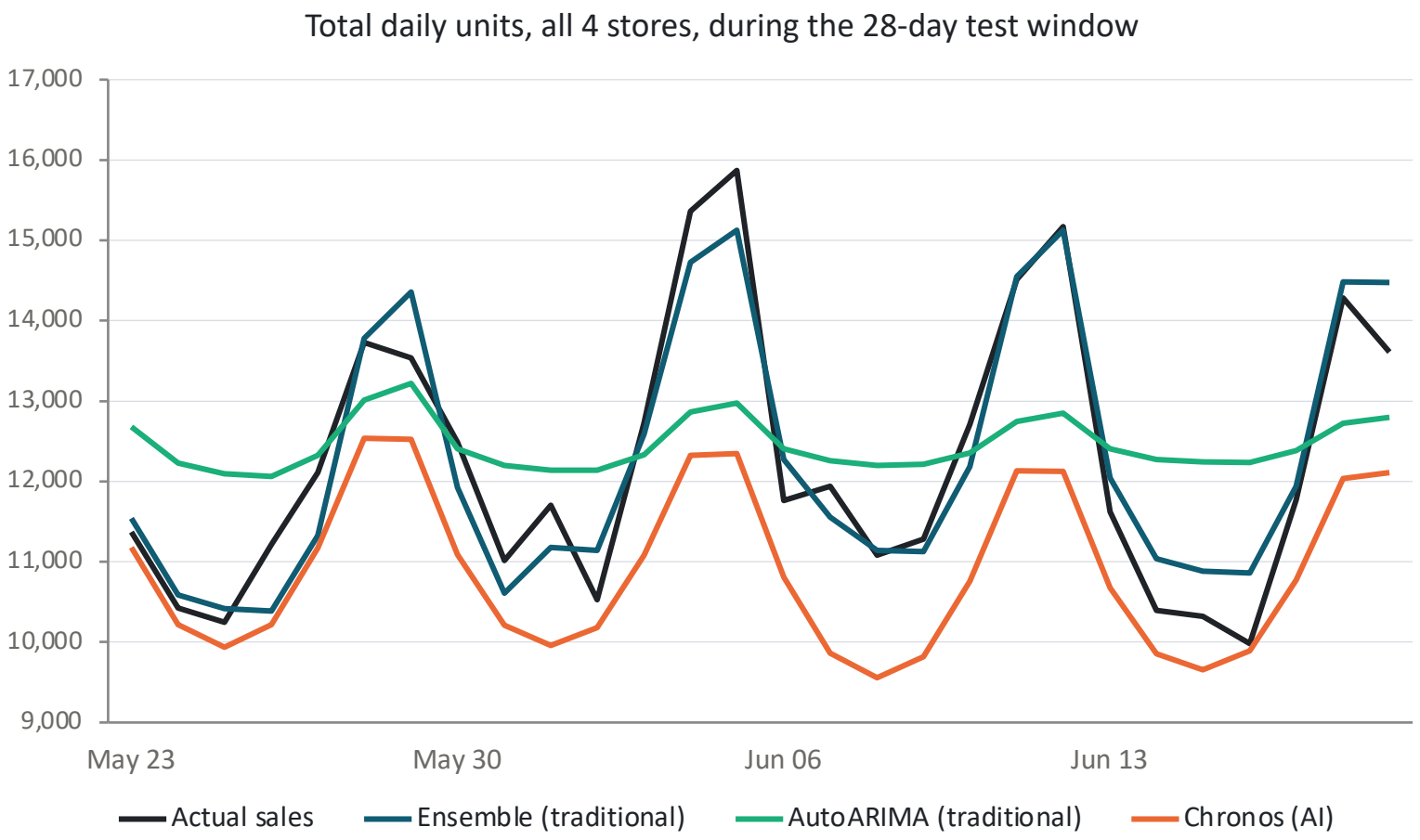
Traditional models win on accuracy

Model	Overall accuracy (WRMSSE)	Store totals (daily error)	Single items (RMSSE)	Time for all series	Training needed
Ensemble (AutoETS + LightGBM)	**0.604**	**3.6%**	**0.90**	~3 min	Yes
LightGBM	0.646	4.1%	0.92	80 s	Yes
AutoETS	0.661	4.9%	0.93	86 s	Yes
AutoARIMA	0.892	9.9%	0.94	39 min	Yes
Chronos (AI)	0.949	10.4%	0.92	**10 s**	**None**
Seasonal naive (benchmark)	0.814	5.5%	1.17	<1 s	None

Green = best in its column (benchmark excluded)

Best traditional choice: the AutoETS + LightGBM average, for accuracy. **AI:** fast, but no clear accuracy edge.
How to Read: RMSSE = error vs a naive "same as yesterday" forecast; WRMSSE = RMSSE weighted by dollar sales.

Traditional tracks the peaks; the AI runs low



Follows the peaks

The traditional average rises and falls with every weekend

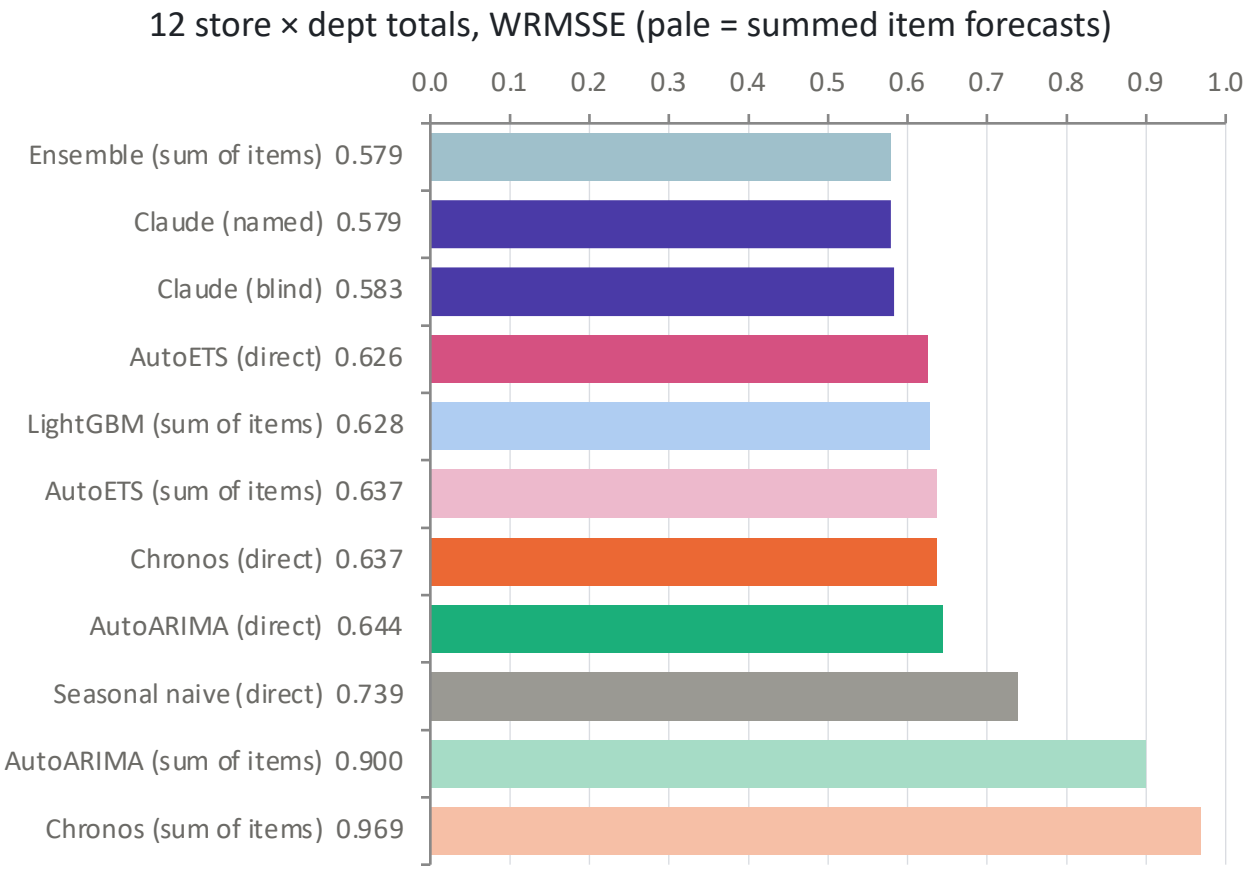
Flat line

AutoARIMA's product-by-product forecasts add up to a flat total

Right shape, too low

Chronos sees the weekly rhythm but forecasts about 11% low

Claude: a language model that reads sales as text



Chronos vs Claude: both pre-trained, zero-shot

	Chronos	Claude
Trained on	Time series	Text
Reads our data as	Numbers	Rows of text
Returns	Quantiles	Forecast + explanation
All 5,748 series	10 s, free	Hours; paid per request

100 sampled items: AutoETS 0.938 best, AutoARIMA 0.951, Claude 0.958. Same items for every model.

Claude covered a subset to keep cost down: 100 random items + 12 totals, compared on the same series. Totals were also fitted directly for the other models.

Using an LLM to forecast: risks and safeguards

Risk	What we did
Pre-training contamination M5 is public and may be in Claude's training data	Named vs blind test: names, dates and year removed. Accuracy unchanged (0.579 vs 0.583)
Unstable output Answers can vary between runs	Fixed prompt and model; every response saved and cached
Cost and scale Paid per request; slow at scale	Tested a subset (100 items + 12 totals); same subset for every model
Malformed answers Wrong length or format	Required a fixed JSON format, checked 28 days returned, retried on failure

The blind test reduces identity cues, but it cannot guarantee that contamination is excluded: the sales pattern itself could be recognised.

Our answer: use a traditional ensemble

Traditional methods lead on accuracy

LightGBM, AutoETS and their average are the three best

AI shows no clear accuracy edge

Chronos trails the benchmark; Claude tested on 112 only

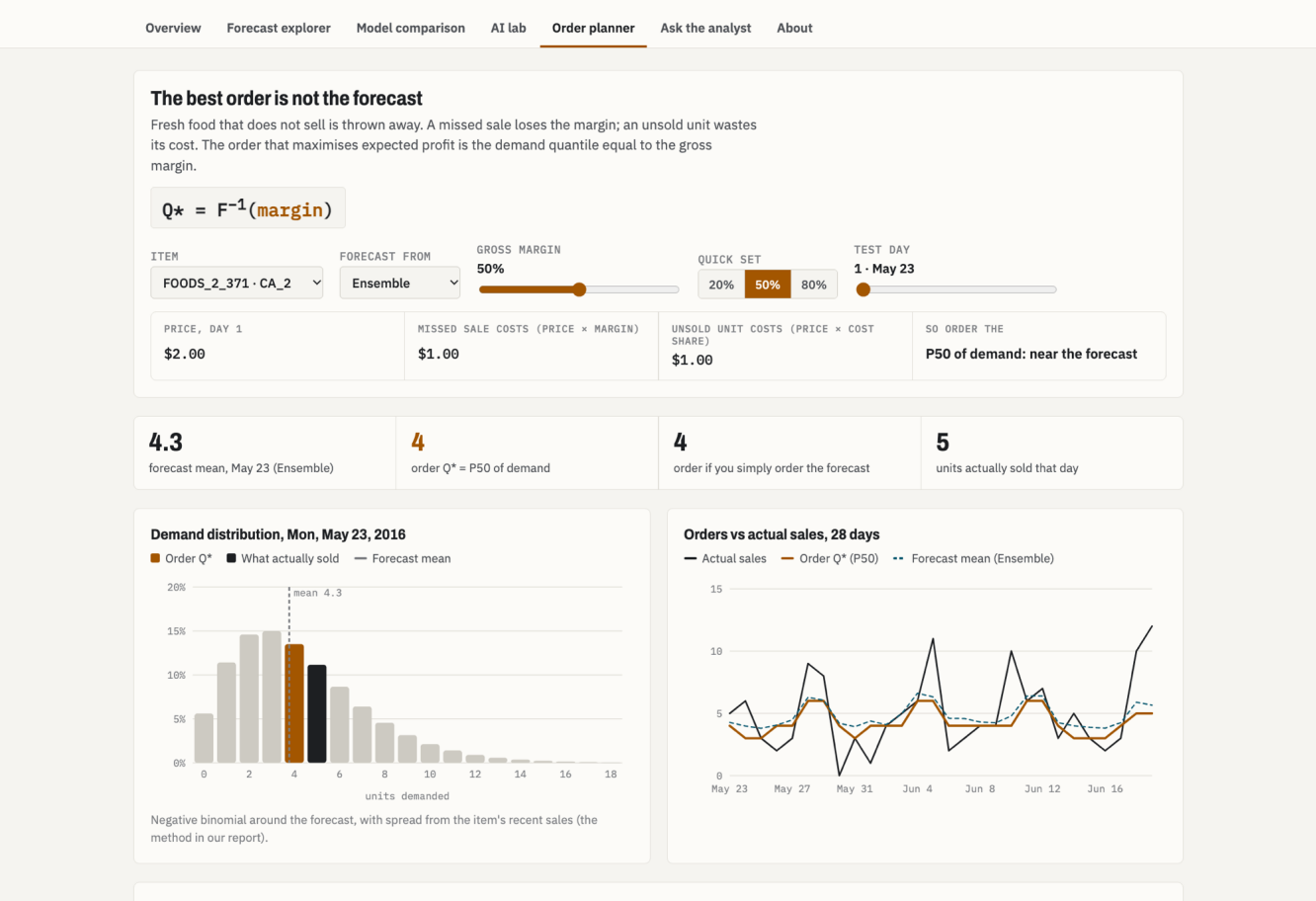
AI lacks interpretability

Black boxes; Claude's reasoning can't be verified

Limitations

- One 28-day test window: close rankings could flip in another month.
- Claude: 112 series only; the blind test can't fully rule out contamination.
- Accuracy is not the business goal. Ordering is a newsvendor problem: running out costs the margin (underage), unsold food costs its price (overage). The best order is a margin-set quantile, not the forecast, and our accuracy metrics weigh both errors equally.

Demo: Shelf Signal dashboard



[Open live dashboard ↗]

Forecast explorer

One store's department: every model against actual sales

AI lab

Claude's own reasoning, and the named-vs-blind test

Order planner

Change the margin: the order moves between P20 and P80

Thank you — questions?